R for Data Science — Final Project Requirements

1. Purpose

The final project is a **group-based data science project** worth **50% of the course grade**. It requires you to apply the full R data science workflow: identify an industry context, build a scientific background from academic literature, obtain a relevant dataset, conduct exploratory data analysis in R, design a dashboard informed by both the literature and the EDA, implement the dashboard as a Shiny application, and document your process and learning in a referenced report.

The project should demonstrate the course learning objectives: tidy and reshape data, summarize and visualize data, integrate external data, build interactive Shiny applications, and communicate insights reproducibly.

2. Project Structure — Required Steps

Step 1: Group Formation and Industry Selection

- Form a group as assigned by the instructor.
- Select an **industry or sector** (e.g., retail, finance, healthcare, education, transportation, energy, public policy, sports, hospitality, or another approved field).
- Define the **business or policy context** clearly: Who is the audience? What decisions could the data inform?
- Submit a short **industry selection note** to the instructor for approval.

Deliverable: Industry selection and problem context (1 page).

Step 2: Scientific Background and Literature Review

- Conduct a **focused literature review** using academic papers, industry reports, and credible sources.
- Identify:
    - The key problem or opportunity in the selected industry.
    - Relevant theories, metrics, or frameworks used in prior work.
    - Research gaps or practical questions your project will address.

- Use a **literature review matrix** or synthesis table to organize sources by theme, method, and relevance.
- Minimum expectation: **5–8 credible academic or industry sources**, properly cited.

Deliverable: Literature review summary and research question(s) (2–3 pages).

Step 3: Dataset Identification and Acquisition

- Find a **real dataset** relevant to your industry and research questions.
- The dataset must be:
    - Accessible and legally usable.
    - Documented with a source, codebook or data dictionary, and collection context.
    - Sufficiently rich for EDA and dashboard development.
- Public datasets are encouraged. If using private or restricted data, obtain instructor approval and ensure ethical handling.
- Import the data into R using reproducible code.

Deliverable: Dataset description, source citation, codebook summary, and import script.

Step 4: Exploratory Data Analysis (EDA) in R

Systematically explore the dataset using R and the tidyverse. Your EDA should include:

- Data cleaning and tidying.
- Handling missing values, duplicates, and inconsistencies.
- Descriptive statistics and summaries.
- Visual exploration using ggplot2 (distributions, relationships, trends, comparisons, anomalies).
- Identification of patterns, outliers, and limitations.
- Initial insights that connect back to your literature review.

Document all code and decisions in an R Markdown or Quarto file.

Deliverable: EDA notebook/report with code, visualizations, and written interpretation.

Step 5: Thought Framework and Dashboard Design

Use the **scientific reading** and **EDA results** to build a thought framework that guides the dashboard design. This framework should specify:

- Target audience and their information needs.
- Key questions the dashboard must answer.
- Metrics, KPIs, and dimensions to display.
- Visual encodings and chart choices.
- Interactivity requirements.
- Narrative flow or user journey through the dashboard.
- How the design is grounded in the literature and EDA findings.

Create a **dashboard design specification** before implementation: wireframes, component list, and rationale.

Deliverable: Thought framework and dashboard design specification.

Step 6: Dashboard Execution as a Team

Build an interactive dashboard in **Shiny** using R. Requirements:

- A working Shiny application with a clear UI and server structure.
- At least **three meaningful interactive components** (e.g., filters, selectors, reactive plots, reactive tables).
- Integration of the cleaned dataset and EDA insights.
- Appropriate layout and visual design for the target audience.
- Reproducible code organized in a project directory.
- The app should run locally from your submitted files. Deployment is encouraged but not required unless specified by the instructor.

Deliverable: Shiny app source code and a short user guide.

Step 7: Final Report with References

Write a structured report documenting each step and the learning gained. The report should include:

1. Title page and group member contributions.
2. Executive summary.
3. Industry context and problem statement.
4. Scientific background and literature review.
5. Dataset description and acquisition.
6. EDA process, findings, and visualizations.
7. Thought framework and dashboard design rationale.
8. Dashboard implementation and demonstration.
9. Discussion of limitations, ethical considerations, and future work.
10. References in a consistent citation style (APA recommended).
11. **Individual reflections** from each group member documenting their learning from each project step.

The report must be reproducible: include or link to R scripts, R Markdown/Quarto files, and the Shiny app.

Deliverable: Final report (PDF or HTML) with references and individual reflections.

Step 8: Final Presentation

- Present the project as a team.
- Duration: **10–15 minutes**, followed by Q&A.
- Include:
    - Industry context and research question.
    - Key literature findings.
    - Dataset and EDA highlights.

- Dashboard design rationale.
- Live or recorded Shiny dashboard demonstration.
- Lessons learned and limitations.

Deliverable: Presentation slides and live/recorded demo.

3. Timeline and Milestones

Session	Milestone	Due / Activity
Session 5	Project overview, academic reading, literature review methodology	Submit **project design plan and timeline**; industry selection approved
Session 6	Data storytelling, project design, code design	Submit **literature review summary and dataset description**; dashboard design specification draft
Session 7	Project development, output review, scientific rigor	Submit **EDA notebook and dashboard prototype** for code/output review
Session 8	Supervised teamwork, final presentation and submission	Submit **final report, Shiny app, presentation, and individual reflections**; present in class

Instructor may adjust dates; check the course schedule for exact deadlines.

4. Assessment Rubric

Component	Weight	Description
Industry context and problem definition	5%	Clear industry, audience, and decision context
Literature review and scientific background	15%	Quality, relevance, synthesis, and citation of sources
Dataset identification and documentation	10%	Relevance, legality, codebook, reproducibility of import
EDA in R	20%	Cleaning, summaries, visualizations, insight generation, code quality

Thought framework and dashboard design	15%	Coherence between literature, EDA, and design rationale
Shiny dashboard execution	20%	Functionality, interactivity, usability, design, reproducibility
Final report and references	10%	Structure, clarity, referencing, reproducibility, individual reflections
Presentation	5%	Communication, demo, teamwork, Q&A

Total: 100%

The final project is **50% of the overall course grade**.

5. Submission Requirements

Submit a single compressed folder or repository link containing:

- Final report (PDF/HTML).
- R Markdown/Quarto source files.
- R scripts for data import, cleaning, EDA, and dashboard.
- Shiny app folder with ui.R, server.R, or app.R.
- Dataset or a clear data-access script/link.
- Presentation slides.
- Individual reflection documents.
- A README.md explaining how to run the project.

Naming convention: GroupNumber_Industry_FinalProject.

6. Academic Integrity and Data Ethics

- All sources must be properly cited.
- Data must be used ethically and in accordance with its license or terms of use.
- Do not use confidential, personally identifiable, or unauthorized data.
- All code must be authored by the group; clearly attribute any external code or templates.

- Plagiarism and unauthorized collaboration will be handled according to university policy.

7. Support and Review

Use Session 7 for **output and code review**. Groups are expected to bring a working prototype and receive feedback on:

- Code quality and reproducibility.
- EDA and visualization choices.
- Dashboard design and narrative.
- Alignment with the literature review and scientific rigor.

The teaching team will provide guidance, but groups are responsible for project management, milestone tracking, and final delivery.